1) prime factorization →
2) Discrete log problem →

RSA

1) (K_{pu}, K_{pr}) Generation —* ①
2) Encrypt the message using K_{pu} — ②
3) Decrypt the cipher text using K_{pr} — ③

* Correctness
* hardness → Knowing K_{pu}, it is hard to compute K_{pr}
→ Knowing K_{pu}, ciphertext, it is hard to recover the plaintext without K_{pr}

RSA

1) Key generation

1.a choose two large prime numbers p, q

$n = pq$

1.b Server compute the $\varphi(n)$ given n.
Compute all the number that are coprime to n and less than n

$\{a \text{ s.t. } a<n, \ GCD(a,n)=1\}$

The number of such coprime number relative to n

$\varphi(n) = |\{a, \text{ s.t. } a<n, GCD(a,n)=1\}|$

$\boxed{\varphi(n) = (p-1)(q-1)}$ → result

Example

$p=3, \ q=5$

$n=15$

$\boxed{\varphi(n) = 8}$

$\{1, 2, 4, 7, 13, 14, 11, 8\}$

So far, n, found $\varphi(n)$ ✓

$\mathbb{Z}_{\varphi(n)} = \{0, 1, 2, 3, \ldots \varphi(n)-1\}$ ✓

pick two number e, and d

$\boxed{ed \bmod \varphi(n) = 1}$ —— *

It is possible to find such two number e and d

Example $\varphi(n) = 8$

$\boxed{\mathbb{Z}_8 = \{0, 1, 2, \ldots 7\}}$

$ed \bmod 8 = 1$

$e=3, \ d=3$ $\quad 9 \bmod 8 = 1$

$e=5, \ d=5$ $\quad 25 \bmod 8 = 1$

Key generation

$K_{pu} = (e, n)$ ✓

$K_{pr} = (d, n)$??

Server → clients (e,n)

(d,n) by itself

Encryption

$M < n$ {plaintext}

$\boxed{C = M^e \bmod n} \longrightarrow * \rightarrow C$

$n = 15$?? $M < 15$

the

$\{0, 1, 2, \ldots 14\}$

0000 0001 0010 ... 1110 – {1111}

Video file

{0 1 0 1 .... 0 1 ....}

M

using C, can an adversary recover M ?

$n \approx 2^{128}$ $\quad 2^{196}$

$\boxed{n \approx 2^{56}}$ ✗

Decryption

$\{e, n, e, d\}$

K_{pr}

$\boxed{M = C^d \bmod n}$ —*

RHS

proof correctness

$C^d \bmod n$

$(M^e \bmod n)^d \bmod n$

$M^{ed} \bmod n$ —*

$ed \bmod \varphi(n) = 1$

$ed = x\varphi(n) + 1$

$M^{ed} \bmod n = M^{x\varphi(n)+1} \bmod n$

$= M^{x\varphi(n)} \cdot M \bmod n$

$= ((M^{x\varphi(n)} \bmod n) M) \bmod n$

Euler's theorem

$\boxed{M^{\varphi(n)} \bmod n = 1}$

M — End of proof

(d,n) Server

$C_1 \leftarrow M_1$ client 1 (e,n)

$C_2 \leftarrow M_2$ client 2 (e,n)

⋮

$C_{1000} \leftarrow M_{1000}$ client 1000 (e,n)

$C_i = M_i^e \bmod n$

$\{C_1, C_2, C_3, \ldots C_{1000}\}$

$C_1^d \bmod n$ → M_1

$C_2^d \bmod n$ → M_2

$C_{1000}^d \bmod n$ → M_{1000}

Example

$n = 187$

$p = 17, \ q = 11$

$\varphi(n) = (p-1)(q-1) = 160$

e and d

$ed \bmod \varphi(n) = 1$

$e=7, \ d=23$

public key = (7, 187)

private key = (23, 187)

client

{0101100 | 1000 | ...... | 101100}

7 bits at a time

$0 \le M \le 127$

$M = 88$

cipher

$C = M^e \bmod n$

$= 88^7 \bmod 187$

$= 11$

$0 \le C \le 186$

179

plaintext → 7 bits

ciphertext → 8 bits (reserve)

ciphertext size ≥ plaintext size

~~AES~~

Decryption

$C = 11$

$C^d \bmod n = 11^{23} \bmod 187$

$= 88$!!!

Audio file → RSA → ciphertext audio file

↓

wifi | 4G | 5G